Motorcycle Modelling in SUMO

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Why Motorcycle Modelling in SUMO Matters

🧠 SUMO models motorcycle geometry, not decision logic.

🏍️ **vClasses:** Set defaults (2.2 × 0.9 m, minGap 2.5 m, minGapLat 0.6 m).

🔀 **Sublane model:** Enables lateral movement (SL2015; Erdmann, 2015).

📏 **Lane filtering:** Emerges indirectly from speed-gain lane changes.

📊 **Key gaps:** No explicit filtering decisions, speed-dependent lateral gaps, staggered following, lean dynamics, or lateral-conflict detection by the SSM safety device.

📚 **Validation evidence:** Kuo et al. (2023) found SUMO and VISSIM weak at motorcycle waiting zones and two-stage left turns, especially with higher PTW volumes.

How SUMO's Sublane Model Works

🎯 SL2015 enables continuous lateral movement; lane-filtering emerges from speed-gain logic, not explicit decision-making.

1. Core Mechanism

- --lateral-resolution <m> activates SL2015 and divides lanes into lateral stripes.
- Vehicles move laterally without overlapping the same stripe longitudinally.
- Use **≤ 0.8 m** for motorcycles; 0.8 m divides a 3.2 m lane evenly.
- Finer resolution improves detail but increases runtime.

2. Decision Flow

- **Car-following:** Any lateral overlap creates a full leader; the minimum safe speed applies.
- **Lateral motivation:** Speed gain, keep-right, alignment and strategic needs influence movement.
- **Lateral maneuver:** Sufficient speed-gain motivation (lcSpeedGain, lcSpeedGainLookahead) triggers movement toward free stripes.
- **Lateral constraints:** Movement is limited by maxSpeedLat, lcAccelLat and minGapLat.

How SUMO's Sublane Model Works (Contd.)

3. Key parameters for motorcycles		
Parameter	**Default**	**Role**
latAlignment	center	Preferred lateral position (left, right, arbitrary, nice, or offset)
minGapLat	0.6 m	Desired lateral gap, constant and independent of speed
maxSpeedLat	1.0 m/s	Maximum lateral speed
lcAccelLat	1.0 m/s^2	Maximum lateral acceleration
lcPushy	0	Willingness to encroach into partly occupied space
lcSublane	1	Eagerness to reach the preferred alignment
lcImpatience / lcTimeToImpatience	0 / ∞	Growing aggressiveness when blocked
lcMaxDistLatStanding	0 for two-wheelers	Prevents sliding sideways while standing, which also limits seepage

4. What it does well, and where it falls short

Works: several motorcycles side by side in one lane, virtual-lane formation, and lateral alignment.

Falls short:

- no explicit gap-seeking filtering or seepage logic
- no speed-dependent lateral clearance
- no lean dynamics
- SSM ignores lateral conflicts

Filtering Criteria and Safety at Signals

🎯 Motorcycle filtering is gap-driven and two-dimensional, requiring surrogate safety measures.

📏 Filtering criteria

- Lee et al. (2009): Filtering, weaving and dynamic virtual lanes.
- Vlahogianni (2014): Lateral cut-offs of 1.2 m (filtering) and 0.9 m (overtaking).
- Promraksa et al. (2022): Clearance varies with speed, filtering side and adjacent vehicle conditions.

🚦 Signalized intersections

- Minh et al. (2012): Queue formation and motorcycle weaving.
- Ameen & Ahmad (2024): 3.4× higher red-light-running likelihood for two-wheelers than car drivers.

⏱️ Surrogate safety measures

- Damani & Vedagiri (2023): Applied TTC and DRAC to filtering.
- SUMO defaults: TTC < 3 s; PET < 2 s; dedicated lateral-gap indicators are lacking.

Calibration Options and Gaps in SUMO

Calibration options

⚙️ **A. Parameters only:** tune lateral-resolution (0.2–0.4 m), minGapLat, lcPushy, jmStoplineGap. Fast, but no explicit seepage or filtering logic.

🔀 **B. TraCI/libsumo:** external gap-seeking filtering via moveToXY/changeSublane.

💻 **C. C++:** new lane-change model derived from SL2015.

↔Gaps to fill

- Speed-dependent lateral gap
- Filtering and seepage at stop lines
- Staggered following
- Lean-limited lateral dynamics
- Lateral-conflict SSM